C-DAC CCEE: Cloud Computing Cheat Sheet

SMCQ Exam Preparation Guide

1. CLOUD FUNDAMENTALS

Definition: On-demand delivery of IT resources over the internet with pay-as-you-go pricing.

NIST 5 Essential Characteristics

(Memorize these for MCQs)

- **On-demand Self-service:** User provisions resources automatically without human interaction.
- **Broad Network Access:** Available over the network on standard devices (mobile, laptop).
- **Resource Pooling:** Multi-tenant model; resources (physical/virtual) are dynamically assigned.
- **Rapid Elasticity:** Scale outward and inward commensurate with demand (often automated).
- **Measured Service:** Resource usage is monitored, controlled, and billed (metering).

Key Benefits

- **Trade CapEx for OpEx:** No huge upfront cost; pay for what you use.
- **Speed & Agility:** Resources in minutes, not weeks.
- **Global Scale:** Deploy apps in multiple regions easily.

2. SERVICE MODELS (SPI)

Who manages what? (Crucial for Exam)

SaaS (Software as a Service)

Provider Manages: Everything (App, Data, Runtime, Middleware, OS, Hardware). **User Manages:** Nothing (just configuration).

- *Examples:* Gmail, Salesforce, Office 365, Zoom.
- *Pros:* No maintenance, accessible anywhere.
- *Cons:* Less control, data privacy concerns.

PaaS (Platform as a Service)

Provider Manages: OS, Middleware, Runtime, Hardware. **User Manages: Applications & Data**.

- *Examples:* Google App Engine, AWS Elastic Beanstalk, Heroku.
- *Target:* Developers (focus on code, not servers).

IaaS (Infrastructure as a Service)

Provider Manages: Virtualization, Servers, Storage, Networking. **User Manages: OS, Middleware, Apps, Data**.

- *Examples:* AWS EC2, Azure VM, Google Compute Engine.
- *Target:* SysAdmins (maximum control).

3. DEPLOYMENT MODELS

- **Public Cloud:** Infrastructure is open to the general public. Owned by provider (AWS, Azure).
- **Private Cloud:** Infrastructure used exclusively by a single organization. High security/control.
- **Hybrid Cloud:** Composition of Public + Private. Enables *Cloud Bursting* (spillover traffic).
- **Community Cloud:** Shared by organizations with common concerns (e.g., banks, hospitals).

4. VIRTUALIZATION

The backbone of Cloud Computing.

Hypervisor (VMM): Software that creates and runs Virtual Machines (VMs).

Types of Hypervisors

Type 1 (Bare Metal)	Type 2 (Hosted)
Runs directly on hardware.	Runs on top of a Host OS.
High performance, Enterprise use.	Slower, used for testing/labs.
Ex: VMware ESXi, Hyper-V, Xen.	*Ex:* Oracle VirtualBox, VMware Workstation.

Key Terms

- **VM Provisioning:** Allocating resources to create a VM.
- **Live Migration:** Moving a running VM between hosts with zero downtime.

5. ECONOMICS & PRICING

CapEx vs. OpEx

- **CapEx (Capital Expenditure):** Upfront cost for physical assets (servers, buildings). Deducted over time.
- **OpEx (Operating Expenditure):** Day-to-day costs (subscriptions, electricity). Pay-as-you-go. Cloud is primarily OpEx.

Pricing Strategies

- **On-Demand:** Pay by hour/second. Flexible, no commitment. Expensive.
- **Reserved Instances:** 1 or 3-year commitment. Significant discount (up to 70%).
- **Spot Instances:** Bid for unused capacity. Cheapest (up to 90% off) but can be interrupted.

6. VENDOR COMPARISON

Memorize these mappings!

Feature	AWS	Azure
Compute	EC2	Virtual Machine
Storage (Object)	S3	Blob Storage
Serverless	Lambda	Azure Functions
Database (SQL)	RDS	Azure SQL DB
Network	VPC	VNet
Identity	IAM	Entra ID (AD)

AWS Specifics

- **Region:** Geographic location (e.g., Mumbai).
- **AZ (Availability Zone):** Isolated data center within a Region.
- **AMI:** Amazon Machine Image (OS template for EC2).

7. EXAM QUICK TIPS

- If the question mentions "Focus on code, not OS" $\rightarrow$ **PaaS**.
- If the question mentions "Legacy app migration" or "Full OS control" $\rightarrow$ **IaaS**.
- If the question mentions "Bare Metal" $\rightarrow$ **Type 1 Hypervisor**.
- **Scalability** = Ability to grow. **Elasticity** = Auto grow/shrink.